

Autonomous Lead Generation & Validation Pipeline

n8n Enterprise Automation Architecture & Technical Specification

ARCHITECTURE TYPE	CORE ENGINE	AI INTEGRATION	TARGET DESTINATION
Multi-Stage ETL & Validation	n8n Workflow Automation	Groq API (LLM Extraction)	Google Sheets (Region Partitioned)

1. Workflow Overview

This technical architecture document outlines the end-to-end execution flow of the automated lead generation and validation system. Designed for high throughput and data integrity, the pipeline extracts raw business intelligence, validates physical location and communication endpoints, scores validity via advanced heuristics, and partitions leads by geographical region into dedicated Google Sheets targets.

n8n Workflow Execution Diagram

Figure 1.1: Complete n8n Workflow Topology showing Schedule Trigger, Data Scraping, Phone/Email Validation, and Regional Google Sheets Routing with compact node representations.

2. Pipeline Execution Stages & Node Specifications

The workflow is structured into sequential processing blocks utilizing compact modular nodes for maximum efficiency and maintainability.

TRIGGER / INPUT Schedule & Config	Phase 1: Initialization & Parameter Definition Schedule Trigger → Define Countries → Give Input Initiates the execution cycle on a scheduled interval. The initialization nodes load target geographical parameters (AE, GB, AU, PK) and runtime configuration schemas dynamically.
EXTRACTION Scraping & Splitting	Phase 2: Raw Data Harvesting & Record Splitting Get Data → Split Out → Number Availability Executes automated web-scraping queries against business directories. The raw batch output is split into discrete operational items via Split Out , followed by initial telecom availability checks.

VALIDATION

Phone, Web & AI

Phase 3: Multi-Layer Validation & Groq AI Processing

Phone Validator → Web Availability → Groq API

Performs deep telephony verification via external APIs ([Veriphone](#)), tests web asset liveness, and routes unstructured payloads through the [Groq API](#) for entity normalization and structured schema mapping.

ROUTING

Email & Partition

Phase 4: Email Verifier & Regional Partitioning

Email Checker → Edit Fields → AE / GB / AU / PK

Validates MX records and inbox deliverability using email checking endpoints. Cleaned records pass through field mapping and are dynamically filtered and routed to region-specific Google Sheets targets ([Sheet 8 to 11](#)).

3. Mathematical Verification Model

Lead scoring and validity filtration within the pipeline adhere to composite probability thresholds defined as:

$$V_{score} = w_1 P_{phone} + w_2 P_{email} + w_3 P_{web} \geq \theta_{thresh}$$

Where P_{phone} , P_{email} , and P_{web} represent validation confidence scores for respective channels, weighted by operational importance factors w_i against strict organizational thresholds θ_{thresh} .